Scaling Bayesian models

Some considerations for larger model fits

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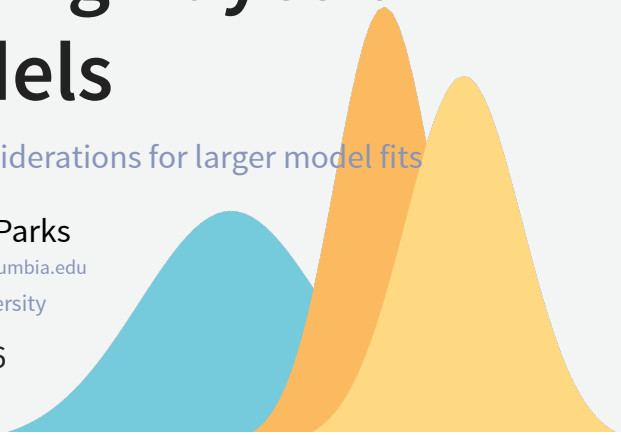


So far…

- Models fitted with MCMC in `NIMBLE`:
  - True sampling of posterior
  - Flexible, but can become slow with larger models
- Models fitted with `R-INLA`:
  - Deterministic (but usually very accurate) approximation
  - Seconds rather than minutes
- Both fitted on a Posit cloud, for 107 Italian provinces in previous sessions in this workshop

This session

- In this session we look at what changes when models and data get much bigger:
  - Optimizing models
  - Running on more powerful computing systems
  - Exploring model results interactively

This session

1. Building up an annual space–time model for Italy from the earlier sessions
2. An example research model at full national scale in the US, and how each component builds on this workshop series
3. Specification choices that keep a large model tractable
4. `INLA` settings that can make large fits faster
5. Scaling even further: what to do when that is not enough
6. Outputs and exploration at scale
7. High-performance computing

Building up a space–time model

From the earlier labs to a space–time model

Before looking at anything at national scale, let's build a full space–time model on data we already know: the Italian provinces.

- Where the `INLA` lab left off: a BYM2 model for the provinces of Italy, for September 2018 only — 107 spatial units, one `f()` term, fitted in seconds
- Where we go now: **all provinces, all years** — an annual space–time model, built up one `f()` term at a time
- This is also the model you will build yourselves in this session's lab

Building the Italy model: the data

Step one is to stop throwing data away — but also to be deliberate about which dimensions we keep. Here we aggregate the months away and model **annual** counts per province:

```
data <- read_rds(here("data", "italy", "italy_mortality.rds")) |>
  group_by(SIGLA, year) |>
  summarise(deaths   = sum(deaths),
            expected = sum(expected), .groups = "drop") |>
  arrange(SIGLA, year)
```


Building the Italy model: the data

`R-INLA` needs a separate index column for every `f()` term, even when several terms index the same thing:

```
data <- data |>
  mutate(
    id_space     = as.integer(factor(SIGLA)),   # BYM2 baseline
    id_year      = as.integer(factor(year)),    # RW2 trend
    id_space_int = id_space,                    # interaction: space index
    id_year_int  = id_year,                     # interaction: time index
    id_cell      = row_number(),                # interaction: one per cell
    id_year_c    = id_year - mean(unique(id_year))  # centred year, for slo
  )
```

- The adjacency graph `g` is rebuilt from the shapefile with `poly2nb()` and `nb2INLA()`
- `id_space_int` and `id_year_int` are duplicates

Building the Italy model: in symbols

$$y_{jt} \sim \text{Pois}(\mu_{jt})$$
$$\log(\mu_{jt}) = \log(E_{jt}) + \alpha + b_j + \gamma_t + \nu_{jt}$$

- b_j: BYM2 spatial baseline — 107 provinces (spatial session)
- γ_t: smooth trend across years, RW2 (non-linear session)
- ν_{jt}: the **space–time interaction** — how each province departs from the common trend

Building the Italy model: baseline

The starting point is the BYM2 model from the INLA lab, unchanged except that it now runs over every province–year:

```r
hyper_pc <- list(prec = list(prior = "pc.prec", param = c(1, 0.01)))

hyper_bym2 <- list(
  prec = list(prior = "pc.prec", param = c(1, 0.01)),
  phi  = list(prior = "pc",      param = c(0.5, 0.5))
)

formula_1 <- deaths ~ 1 +
  f(id_space, model = "bym2", graph = g, scale.model = TRUE,
    hyper = hyper_bym2)
```

With only this term, every year gets the same map — the model cannot describe change over time at all.

Building the Italy model: the common trend

Add the global temporal term: a smooth trend across years, shared by every province.

```r
formula_2 <- deaths ~ 1 +
  f(id_space, model = "bym2", graph = g, scale.model = TRUE,
    hyper = hyper_bym2) +
  f(id_year,  model = "rw2", constr = TRUE, scale.model = TRUE,
    hyper = hyper_pc)
```

- RW2 gives a smooth, non-linear national trend — the smoothing idea from the non-linear regression session, as a random walk rather than a spline basis
- `constr = TRUE` (sum-to-zero) keeps the trend identifiable alongside the intercept

Building the Italy model: the common trend

Add the global temporal term: a smooth trend across years, shared by every province.

```r
formula_2 <- deaths ~ 1 +
  f(id_space, model = "bym2", graph = g, scale.model = TRUE,
    hyper = hyper_bym2) +
  f(id_year,  model = "rw2", constr = TRUE, scale.model = TRUE,
    hyper = hyper_pc)
```

Now every province rises and falls **in exactly the same way**: the maps for different years are identical up to a single common shift.

Space-time interaction

The separable model says: province $_j$ has its own level, year $_t$ has its own level, and that is all.

- Real data are rarely so obliging — provinces diverge, some improve faster than others, some years hit some regions harder
- Anything the separable model cannot express ends up in the residuals

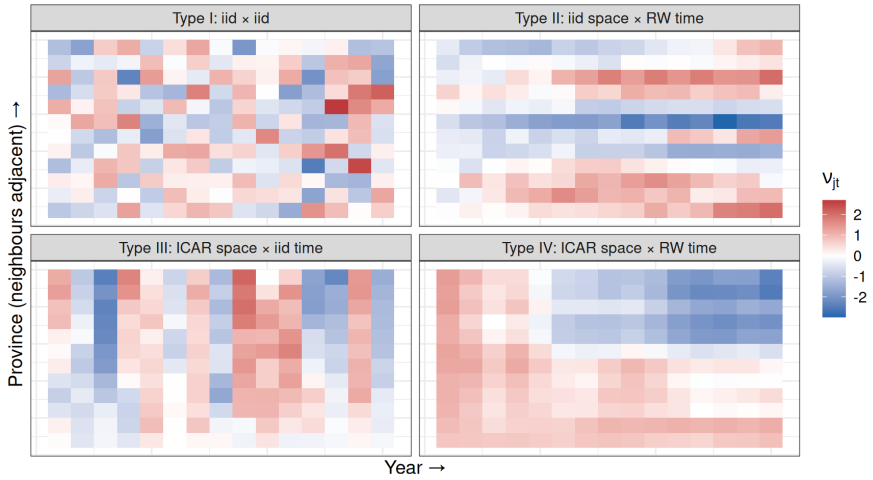
Space-time interaction

The fix is the **interaction** term $_{v_{jt}}$, and in the spatiotemporal session we saw four ways to structure it (Knorr-Held, 2000):

$$v_{jt}: \quad \theta_j \otimes \gamma_t \mid \theta_j \otimes \xi_t \mid \phi_j \otimes \gamma_t \mid \phi_j \otimes \xi_t$$

Each combines an unstructured ($_\theta$, $_\gamma$) or structured ($_\phi$ spatial, $_\xi$ temporal) effect on each side.

The four types (I, II, III, IV)



Type I: unstructured × unstructured

$$v_{jt} = \theta_j \otimes \gamma_t, \qquad v_{jt} \overset{iid}{\sim} N(0, \sigma_v^2)$$

Departures have **no pattern at all** — pure province–year noise, unrelated to neighbouring provinces or adjacent years.

```
hyper_int <- list(prec = list(prior = "pc.prec", param = c(0.5, 0.01)))

formula_I <- deaths ~ 1 +
  f(id_space, model = "bym2", graph = g, scale.model = TRUE,
    hyper = hyper_bym2) +
  f(id_year,  model = "rw2", constr = TRUE, scale.model = TRUE,
    hyper = hyper_pc) +
  f(id_cell,  model = "iid", hyper = hyper_int)      # <- the interaction
```

- The cheapest structure to fit: no graph, no ordering, one precision
- Useful as a catch-all for overdispersion beyond the main effects, but it borrows no strength — a sparse cell is shrunk towards zero, not towards its neighbours

Latent values: $N_p \times N_t = 107 \times N_t$. Hyperparameters: 1.

Type II: unstructured space × structured time

$$\nu_{jt} = \theta_j \otimes \xi_t, \qquad \nu_{j\cdot} \sim \text{RW1 independently for each } j$$

Each province gets its own smooth trend, but the trends are unrelated across the map: Milan's departure evolves smoothly year to year, and tells you nothing about Bergamo's.

```
formula_II <- deaths ~ 1 +
  f(id_space, model = "bym2", graph = g, scale.model = TRUE,
    hyper = hyper_bym2) +
  f(id_year,  model = "rw2", constr = TRUE, scale.model = TRUE,
    hyper = hyper_pc) +
  f(id_space_int, model = "iid", hyper = hyper_int,  # <- the interaction
    group = id_year_int,
    control.group = list(model = "rw1", scale.model = TRUE))
```

- The `group =` Kronecker construction: IID over provinces, RW1 across the grouped years
- This is the fully flexible version of the linear slopes we come to shortly

Latent values: $N_p \times N_t$. Hyperparameters: 1.

Type III: structured space × unstructured time

$$\nu_{jt} = \phi_j \otimes \gamma_t, \qquad \nu_{\cdot t} \sim \text{ICAR}(W) \text{ independently for each } t$$

Each year gets its own spatial pattern — smooth across the map within a year, but unrelated to the year before or after. Natural for events that strike different regions in different years.

```
formula_III <- deaths ~ 1 +
  f(id_space, model = "bym2", graph = g, scale.model = TRUE,
    hyper = hyper_bym2) +
  f(id_year,  model = "rw2", constr = TRUE, scale.model = TRUE,
    hyper = hyper_pc) +
  f(id_space_int, model = "besag", graph = g,        # <- the interaction
    scale.model = TRUE, hyper = hyper_int,
    group = id_year_int,
    control.group = list(model = "iid"))
```

- The same construction as Type II with the roles swapped: Besag over provinces, IID across the grouped years

Latent values: $N_p \times N_t$. Hyperparameters: 1.

Type IV: structured × structured

$$\nu_{jt} = \phi_j \otimes \xi_t, \qquad \nu \sim \text{ICAR}(W) \otimes \text{RW1}$$

Departures evolve **smoothly in both dimensions at once**: a province's excess resembles its neighbours' *and* its own value last year. The most realistic structure, and the hardest to identify.

```
formula_IV <- deaths ~ 1 +
  f(id_space, model = "bym2", graph = g, scale.model = TRUE,
    hyper = hyper_bym2) +
  f(id_year,  model = "rw2", constr = TRUE, scale.model = TRUE,
    hyper = hyper_pc) +
  f(id_space_int, model = "besag", graph = g,        # <- the interaction
    scale.model = TRUE, hyper = hyper_int,
    group = id_year_int,
    control.group = list(model = "rw1", scale.model = TRUE))
```

- Type IV needs extra sum-to-zero constraints beyond `INLA`'s defaults to be identifiable alongside the main effects (via `extraconstr`) — worth knowing before using it in earnest

Latent values: $N_p \times N_t$. Hyperparameters: 1 (plus constraint bookkeeping).

Comparing the four

Type	Space	Time	Borrows strength across	Typical use
I	IID	IID	everything	catch-all
II	IID	RW	years, within province	province-specific trends
III	ICAR	IID	provinces, within year	year-specific maps
IV	ICAR	RW	both	genuinely evolving spatial patterns

- All four cost the **same** $N_p \times N_t$ latent values and one hyperparameter — the difference is what they smooth, not how large they are
- In the South African cervical cancer example from the spatiotemporal session, that was 9 provinces × a handful of years, and a full type IV was comfortably fittable
- For Italy, $107 \times N_t$ is still potentially fittable — which is why the lab uses this data to demonstrate what happens next

Summary of additional model terms

- An annual space–time model for Italy: three `f()` terms, a handful of hyperparameters, and it fits on a laptop in minutes
- Every term came from an earlier session: BYM2 (spatial), RW2 smoothing (non-linear regression), partial pooling (hierarchical), PC priors (INLA)
- We will revisit during the lab

Considerations for a model at even larger scale

A United States model (real example)

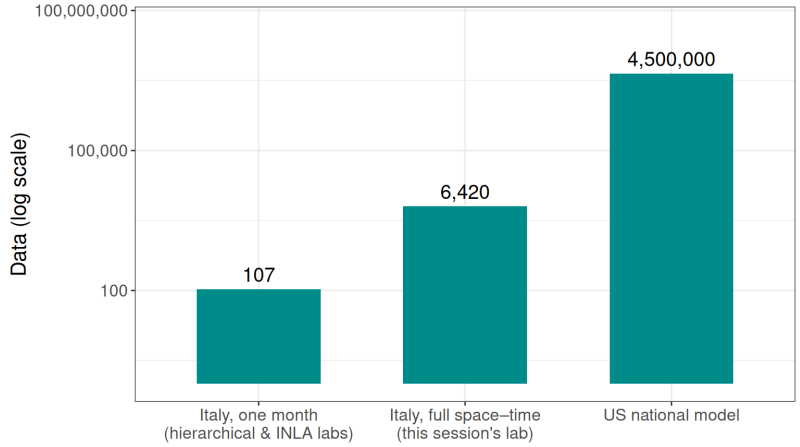
Now the entire United States at county level:

Dimension	Size
Counties	~3,100
Years	~20
Age groups	9
Race/ethnicity groups	4+
Sexes	2

That is around ~4.5 million rows, a spatial graph with thousands of nodes, and a dozen `f()` terms in `R-INLA`, each with its own hyperparameters.

Model size progression

How far this is from where the workshop started:



Why this is a challenge to fit

- **Memory**: the sparse precision matrix of the latent field, its Cholesky factor, and every posterior marginal all sit in RAM. Large spatiotemporal `INLA` models can need hundreds of GB.
- **Time**: hours rather than seconds, and a sensitivity analysis means refitting several times over.
- **Typical laptop**: ~16–64 GB of memory.

The model, term by term

The US model: data and indices

The same pattern as the Italy model — one index column per `f()` term, plus the weight variable for the slopes:

```
dat <- readRDS(file.path(adrd_data_prep, "dat_inla_ready.rds")) |>
  filter(sex == "F")

g <- inla.read.graph(filename = paste0(data.folder, "shapefiles/map.adj"))
```

Index	Used by
`id_year`, `id_county`, `id_age`, `id_race`	the four main effects
`id_county2`, `id_race2`, `id_age2`	linear time-trend interactions
`id_county3`, `id_race3`	race-specific spatial fields
`id_county_as1` … `id_county_as9`	age-specific spatial fields
`id_age4`, `id_race4`	race-specific age curves
`id_year_c`	centred year — the weight for every linear slope

The US model: priors

Three tiers, by the role each term plays:

```
# Weakly informative: RW, IID and group terms
hyper_lg    <- list(prec = list(prior = "loggamma", param = c(1, 0.001)))

# Main effect — genuine large between-county variation expected
hyper_bym2  <- list(prec = list(prior = "pc.prec", param = c(1, 0.01)),
                    phi  = list(prior = "pc",      param = c(0.5, 2/3)))

# Interactions — deviations from main effects, so should be smaller
hyper_besag <- list(prec = list(prior = "pc.prec", param = c(0.5, 0.01)))
```

We come back to why the tiering matters for scaling later in the session.

Step 1: main effects

```
model_formula <- deaths ~ 1 +

  # Global temporal trend (RW2)
  f(id_year,   model = "rw2", constr = TRUE, scale.model = TRUE,
    hyper = hyper_lg) +

  # County baseline: spatial + unstructured (BYM2)
  f(id_county, model = "bym2", graph = g, scale.model = TRUE,
    hyper = hyper_bym2) +

  # Global age curve (RW1)
  f(id_age,    model = "rw1", constr = TRUE, scale.model = TRUE,
    hyper = hyper_lg) +

  # Race/ethnicity main effect (IID)
  f(id_race,   model = "iid", hyper = hyper_lg)
```

Fitted with `family = "nbinomial"` and `E = dat$population`.

Step 1: main effects

- The BYM2 county term is exactly the model from the spatial session — the Swiss childhood cancer example — at ~3,100 areas instead of 26 cantons
- The RW2 year term is the smooth trend from the non-linear regression session, as a random walk rather than a spline basis
- `constr = TRUE` (sum-to-zero) keeps each term identifiable alongside the intercept
- `scale.model = TRUE` makes the precisions comparable across structured terms — and makes the PC priors mean what we think they mean

As in Italy, this is a separable model: every county follows the same national trend, every race group the same age curve.

Step 2: linear time-trend interactions

```
model_formula <- deaths ~ 1 +
  # ... four main effects as above ... +

  # ST_lin: county-specific linear trends, spatially smoothed (Besag)
  f(id_county2, id_year_c, model = "besag", graph = g,
    scale.model = TRUE, hyper = hyper_besag) +

  # RT_lin: race-specific linear trends (IID across race)
  f(id_race2,   id_year_c, model = "iid", hyper = hyper_lg) +

  # AT_lin: age-specific linear trends (RW1, smoothed across age)
  f(id_age2,    id_year_c, model = "rw1", constr = TRUE,
    scale.model = TRUE, hyper = hyper_lg)
```

All three take `id_year_c` as a **weight** — the same linear-slope device we just used for Italy, applied on three different dimensions.

Step 2: linear time-trend interactions

- `f(id_county2, id_year_c, model = "besag")` is precisely the Italy interaction, at 3,100 counties
- `f(id_race2, id_year_c, model = "iid")` is its unstructured cousin — the type II analogue, restricted to a line
- `f(id_age2, id_year_c, model = "rw1")` smooths the slopes across *age groups*: adjacent ages should be trending similarly

Step 3: race-specific spatial fields

Four maps, one per race/ethnicity group — but sharing one spatial precision:

```
model_formula <- deaths ~ 1 +
  # ... main effects and linear trends as above ... +

  # RS: race-specific spatial fields (Besag x IID)
  f(id_county3, model = "besag", graph = g, scale.model = TRUE,
    hyper = hyper_besag,
    group = id_race3,
    control.group = list(model = "iid", hyper = hyper_lg))
```

- The `group =` Kronecker construction from the Italy build — Besag over counties, IID between race groups
- One precision estimated for all four maps, rather than four estimated separately

Step 4: age-specific spatial fields

Nine maps, one per age group — but only one is estimated:

```
model_formula <- deaths ~ 1 +
  # ... everything above ... +

  # AS: base field is age group 9 (85+), the highest death count
  f(id_county_as9, model = "besag", graph = g, scale.model = TRUE,
    hyper = hyper_besag) +
  f(id_county_as1, copy = "id_county_as9", fixed = FALSE) +   # 45–49
  f(id_county_as2, copy = "id_county_as9", fixed = FALSE) +   # 50–54
  f(id_county_as3, copy = "id_county_as9", fixed = FALSE) +   # 55–59
  # ... as4 through as8 ...
```

- `copy =` re-uses the base field, scaled by an estimated constant (`fixed = FALSE`)
- The assumption: one shared spatial *shape*, different *amplitudes* by age

Step 5: race-specific age curves

```
model_formula <- deaths ~ 1 +
  # ... everything above ... +

  # RA: race-specific age curves (RW1 x IID)
  f(id_age4, model = "rw1", constr = TRUE, scale.model = TRUE,
    hyper = hyper_lg,
    group = id_race4,
    control.group = list(model = "iid", hyper = hyper_lg))
```

- The same `group =` device as step 3, on a different pair of dimensions: a smooth age curve per race group, with the curves tied together by an estimated between-group variance
- Partial pooling from the hierarchical session — applied to entire curves rather than single parameters

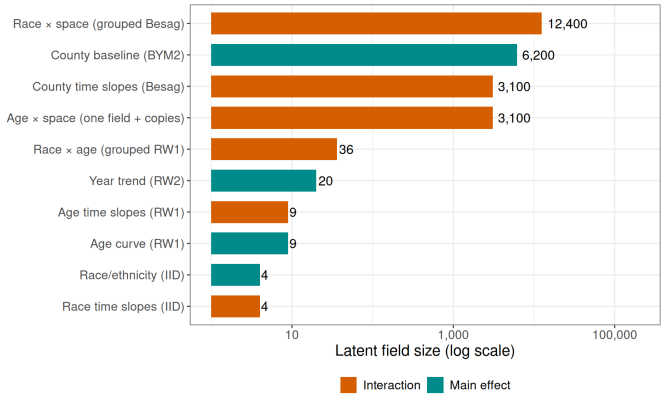
Where each term comes from

Formula term	Where we covered it before
`f(id_race, model = "iid")` — partial pooling	Hierarchical modelling
`f(id_year, model = "rw2")`, `f(id_age, "rw1")`	Non-linear regression; INLA
`f(id_county, model = "bym2", graph = g)`	Spatial modelling
`f(id_county2, id_year_c, model = "besag")`	Spatial modelling; Italy build
`hyper = list(prec = list(prior = "pc.prec", ...))`	INLA

The whole model (real example)

```
model_formula <- deaths ~ 1 +

  # ── Main effects ──────────────────────────────────────────
  f(id_year,    model = "rw2",  constr = TRUE, scale.model = TRUE, hyper = hyper_lg) +
  f(id_county,  model = "bym2", graph = g,     scale.model = TRUE, hyper = hyper_bym2) +
  f(id_age,     model = "rw1",  constr = TRUE, scale.model = TRUE, hyper = hyper_lg) +
  f(id_race,    model = "iid",  hyper = hyper_lg) +

  # ── Linear time-trend interactions (weights = centred year) ────
  f(id_county2, id_year_c, model = "besag", graph = g,
    scale.model = TRUE, hyper = hyper_besag) +
  f(id_race2,   id_year_c, model = "iid",  hyper = hyper_lg) +
  f(id_age2,    id_year_c, model = "rw1",  constr = TRUE,
    scale.model = TRUE, hyper = hyper_lg) +

  # ── Race-specific spatial fields (group =) ─────────────────────
  f(id_county3, model = "besag", graph = g, scale.model = TRUE, hyper = hyper_besag,
    group = id_race3, control.group = list(model = "iid", hyper = hyper_lg)) +

  # ── Age-specific spatial fields (copy =) ───────────────────────
  f(id_county_as9, model = "besag", graph = g, scale.model = TRUE, hyper = hyper_besag) +
  f(id_county_as1, copy = "id_county_as9", fixed = FALSE) +   # 45–49
  f(id_county_as2, copy = "id_county_as9", fixed = FALSE) +   # 50–54
  f(id_county_as3, copy = "id_county_as9", fixed = FALSE) +   # 55–59
```

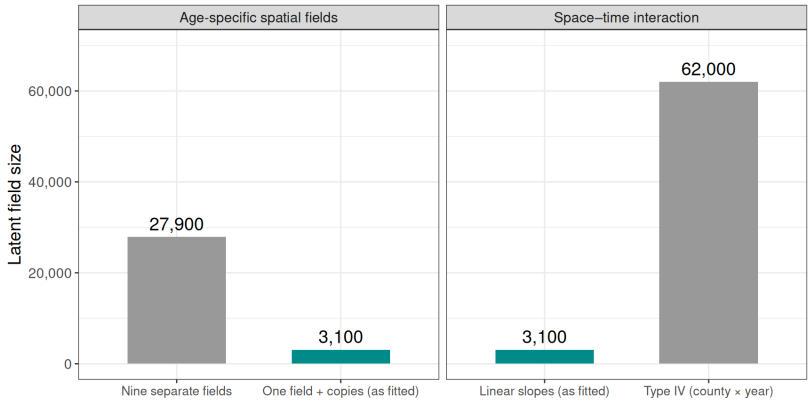

How many parameters is that?



Around 25,000 latent parameters in total — and the interaction terms dominate.

Parameter compromises

The totals only stay at 25,000 because of how the interactions are specified:



The copy feature

Age-specific spatial patterns — nine maps, one per age group:

```
# Base field: the age group with the most deaths (85+)
f(id_county_as9, model = "besag", graph = g, scale.model = TRUE,
  hyper = hyper_besag) +
f(id_county_as1, copy = "id_county_as9", fixed = FALSE) +  # 45–49
f(id_county_as2, copy = "id_county_as9", fixed = FALSE) +  # 50–54
...
```

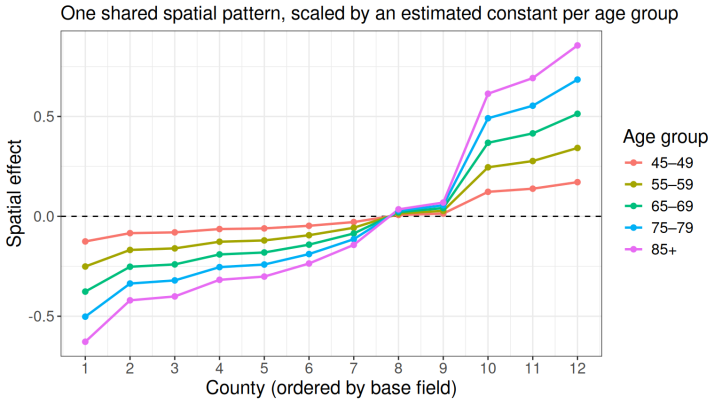
- One spatial field is estimated properly, anchored to the age group with the most data
- The other eight age groups get a **scaled copy**: the same map, multiplied by an estimated constant (`fixed = FALSE`)
- Use with caution

The copy feature

Specification	Latent field	Hyperparameters
Nine independent Besag fields	9 × 3,100	9 precisions
One field + eight copies	3,100	1 precision + 8 scaling constants

- The assumption: the *shape* of the age-specific spatial pattern is shared, and only its *amplitude* varies by age
- Use with caution

The copy feature



This is the same idea as the *spline-by-constant* DLNM from the advanced models session: one shape, different amplitudes.

The copy feature

- Where the shared-shape assumption fails, it fails visibly: plot the base field against crude age-specific SMRs before committing to it
- `group =` and `copy =` are the two main ways R-INLA lets you share structure across strata instead of re-estimating it
- Both trade a modelling assumption for a large reduction in cost — and both borrow strength for the sparse strata

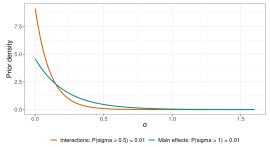
Priors as part of scaling

The big model uses PC priors (INLA session) in deliberate tiers:

```
# Main effect: county baseline (BYM2)
# genuine large between-county variation expected
hyper_bym2  <- list(prec = list(prior = "pc.prec", param = c(1, 0.01)),
                    phi  = list(prior = "pc",      param = c(0.5, 2/3)))

# Interaction terms (Besag slopes, race-spatial fields)
# deviations from main effects, so should be smaller
hyper_besag <- list(prec = list(prior = "pc.prec", param = c(0.5, 0.01)))
```

- $P(\sigma > 1) = 0.01$ for the main effect; $P(\sigma > 0.5) = 0.01$ for the interactions
- Read exactly as in the INLA session — only now the choice of U differs by the *role* of the term

Priors as part of scaling



- PC priors shrink towards the simpler base model ($\sigma = 0$), so interaction terms the data do not support collapse towards zero rather than soaking up noise
- With a dozen variance components this shrinkage is doing real work: it is what stops a richly parameterised model from overfitting — and a well-shrunk model is also better behaved numerically

Making INLA run faster

Approximation settings

`INLA` lets you trade accuracy against cost, via `control.inla`.

Approximation strategy for the latent field:

`strategy=`	Accuracy	Cost
`"gaussian"`	roughest	cheapest
`"simplified.laplace"`	default	moderate
`"laplace"`	best	most expensive

Approximation settings

Integration over the hyperparameters:

`int.strategy=`	What it does
`"eb"`	empirical Bayes: plug in the mode, no integration
`"auto"` / `"grid"`	integrate over hyperparameter uncertainty

The two-stage rough/full workflow

Stage 1 (rough): locate the hyperparameter mode as cheaply as possible.

```
fit_rough <- inla(model_formula, ...,
  control.predictor = list(compute = FALSE),
  control.inla    = list(strategy = "gaussian", int.strategy = "eb"),
  control.compute = list(dic = FALSE, config = FALSE))
```

Stage 2 (full): warm-start from the rough mode, with the accurate settings and the outputs you need.

```
fit <- inla(model_formula, ...,
  control.mode    = list(theta = fit_rough$mode$theta, restart = TRUE),
  control.inla    = list(strategy = "laplace", int.strategy = "auto"),
  control.compute = list(dic = TRUE, config = TRUE))
```


Why this works

- The expensive part of a large `INLA` fit is the numerical optimisation over the hyperparameters: each candidate value of θ requires factorising a large sparse matrix
- The *location* of the mode barely depends on which approximation strategy is used
- So the mode can be found cheaply, and the expensive settings used for a single pass from there

Only compute what you need

Every quantity requested in an `inla()` call costs time and memory, and at scale those costs add up.

- `control.predictor = list(compute = FALSE)` unless you need fitted values from this particular fit
- `dic`, `waic`, `cpo`: only in the fits where you will actually use them
- `config = TRUE` (needed for posterior sampling) only in the final fit
- `keep = FALSE` so the working files are not retained

Threads and solvers

- `num.threads` in `inla()` controls parallelism within a fit; useful, but with diminishing returns, and more threads mean more memory in use at once
- The PARDISO sparse solver can speed up the factorisations considerably on large models — see `inla.pardiso()` for setup on your system
- Recent versions of `R-INLA` also include a reworked, lower-memory computational core — worth keeping the package up to date if you are fitting large models

Scaling even further

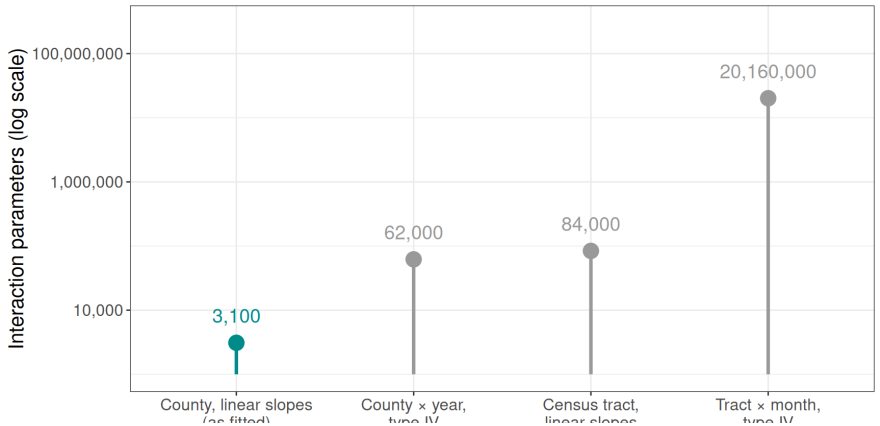
What happens if the model grows again

Each of these is a real request that comes up in practice:

Change	What it multiplies
Annual → monthly time	time points ×12; interaction terms ×12
County → census tract	graph much larger
Adding a stratifier (e.g. education)	rows ×levels; possibly new `f()` terms
Linear → non-linear space–time interaction	interaction parameters ×(time points)

What happens if the model grows again

Just the space–time interaction term, under different choices:



The options, roughly in order

1. **Restrict the interaction structure**: linear slopes, `group =`, `copy =` — as in the model we just walked through
2. **Stratify into independent models**: Smaller models, run in parallel, and simpler to debug than one enormous joint model
3. **Aggregate a dimension you do not need**: if the question is about counties and years, do not carry months
4. **Reduce outputs**: marginals only for the quantities you will report
5. Only then: **bigger hardware**

The options, roughly in order

A note on stratifying (option 2):

- Splitting into independent models has a statistical cost — no borrowing of strength across the strata you split on
- So split on the dimensions where pooling was never intended (sex, cause definition), not the ones where it matters if you can help it (age, race, space)
- The bonus: independent fits can run simultaneously, which is what the job arrays at the end of this session are for

Outputs at scale

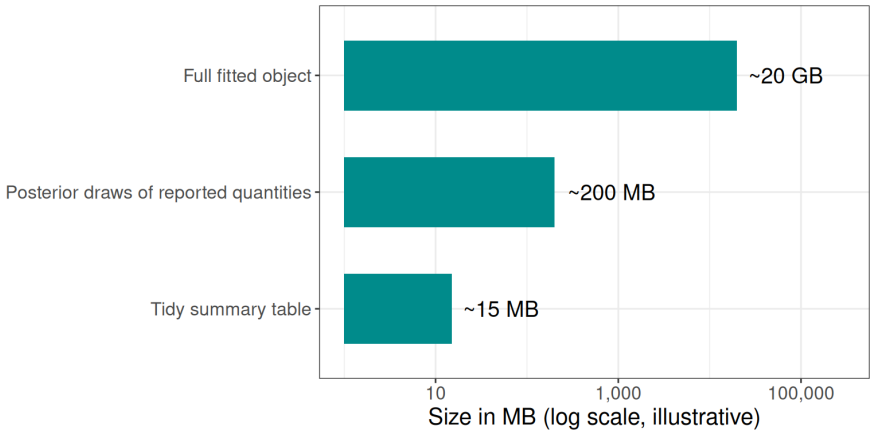
What to save

A full fitted object from a national-scale model can run to tens of GB.

Save instead:

1. **Tidy summary tables**: one row per county × year × group, with the posterior median and 95% credible interval
2. **Posterior samples of the quantities you report**, via `inla.posterior.sample()` (this is what `config = TRUE` is for), so that derived quantities carry proper uncertainty
3. **Hyperparameter summaries** and model metadata: the formula, priors, date, and software versions used

What to save



High-performance computing

High-performance computing, in brief

Once the model is as lean as it can be, some fits genuinely need hundreds of GB of memory — and that means a high-performance computing (HPC) cluster.

- You connect to a **login node**, and submit jobs to a **scheduler** (usually SLURM), which runs them on large **compute nodes**
- A job is just a shell script with a resource request at the top, running the same R script you developed locally:

```
#SBATCH --time=12:00:00
#SBATCH --mem=700G          # a real request, for the model above
#SBATCH --array=1-4         # one job per sex x cause stratum

Rscript fit_stratum.R ${SLURM_ARRAY_TASK_ID}
```


High-performance computing, in brief

- The `--array` line runs the four stratified models simultaneously — the payoff for stratifying into independent fits
- Request resources honestly: too little and the job is killed; far too much and you wait much longer in the queue
- A sensible approach: run a scaled-down version first, check what it actually used (`seff <jobid>`), then request somewhat above that

Reproducibility on a cluster

- The cluster runs a different Linux from your laptop, and `INLA` and `sf` depend on system libraries
- A **container** (Apptainer on most clusters) solves this: a single file holding the operating system, R, and every package at fixed versions

```
apptainer exec --cleanenv --bind "$PWD":"$PWD" \
  ~/r_spatial.sif Rscript fit_stratum.R 1
```

- The same container file gives the same results on the cluster now and on any machine years later — archive it alongside the paper

Getting access to serious computing

Worth investigating early in a project, not when the first fit fails:

- **Your institution's HPC cluster**: most universities run one; access is often free or cheap for researchers, sometimes with priority or extra storage available to purchase. Start with your research computing group
- **National allocations**: in the US, ACCESS (formerly XSEDE) awards free compute time on national systems through a lightweight application; many other countries have equivalents

Getting access to serious computing

- **Cloud computing**: AWS, GCP and Azure rent high-memory machines by the hour — expensive for sustained use, but reasonable for occasional very large fits, and no queue
- **Budget compute into grants**: cluster allocations, cloud credits, or a share of a departmental high-memory node are all legitimate, and increasingly expected, direct costs

For the models in this session, the single most useful specification to ask about is **high-memory nodes** (512 GB–1 TB) — for large `INLA` fits, memory matters much more than core count.

Wrapping up

The approach, end to end

1. **Specify for scale**: restricted interactions, shared structure via `group =` and `copy =`, tiered PC priors, as few hyperparameters as the science allows
2. **Develop small, locally**: subset the data, `strategy = "gaussian"`, iterate quickly
3. **Stage the full fit**: rough then full, warm-started, computing only what is needed
4. **Stratify** into independent fits where pooling was never intended, and run them in parallel
5. **Save summaries and posterior samples**, date-stamped, rather than whole fitted objects
6. **Explore** interactively; **report** the slices that matter
7. Move to HPC only for the fits that genuinely need it — and know how to get access before you do

Getting ready for the lab

The lab for this session

- This lab scales the Italian mortality data up from a single month to the full annual space–time dataset.
- During this lab session, we will:
  1. Build an annual space–time model up term by term — BYM2 baseline, RW2 trend, then the interaction — comparing the Knorr-Held types against Besag linear slopes;
  2. Compare a fit with everything computed against a lean fit, and time the rough/full two-stage workflow;
  3. Turn the model into an `Rscript` and `sbatch` pair, with a job array for the prior sensitivity analysis;

Questions?